MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR,
TRANSISTOR ARRAYS, MONOLITHIC SILICON

This specification is approved for use by Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, transistor arrays. Three product assurance classes and a choice of case outline and lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/108	01	B	C	C
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit
01	Two isolated NPN transistors and one NPN Darlington connected pair, general purpose.
02	Three isolated NPN transistors and one NPN differentially connected pair, general purpose.

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	MIL-M-38510, appendix C, case outline
A	F-1 (14-lead, 1/4" x 1/4", flat-pack)
C	D-1 (14-lead, 1/4" x 3/4", dual-in-line)
D	F-2 (14-lead, 1/4" x 3/8", flat-pack)
M	See figure 3 (12-lead can)

1.2.4 Absolute maximum ratings. 1/

Collector - base voltage 2/	40 Vdc
Collector - emitter voltage 2/	15 Vdc
Collector - substrate voltage 3/	60 Vdc
Emitter - base voltage 2/	5 Vdc
Power dissipation 2/	300 mW
Collector current 2/	50 mA
Storage temperature range	-65°C to +150°C
Junction temperature	175°C
Lead temperature (soldering, 60 seconds)	300°C

1/ The collector of each transistor is isolated from the substrate by an integral diode. The substrate must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action.
2/ Rating applies to each transistor within the array.
3/ Does not apply to Q_5 of device type 02, refer to V_{CEO} rating.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Device type	Limits 1/ Min	Limits 1/ Max	Units
Breakdown voltage, collector to base	$V_{(BR)CBO}$	I_C = 10 µA, I_E = 0	01, 02	40	-	V
Breakdown voltage, collector to emitter	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	01, 02	15	-	V
Breakdown voltage, collector to substrate 2/	$V_{(BR)CUO}$	I_C = 10 µA	01, 02	60	-	V
Breakdown voltage, emitter to base	$V_{(BR)EBO}$	I_E = 10 µA, I_C = 0	01, 02	5.0	-	V
Collector to base cutoff current	I_{CBO}	V_{CB} = 35 V, I_E = 0; $-55°C \le T_A \le +25°C$	01, 02	-	10	nA
		T_A = +125°C		-	0.2	µA
Collector to emitter cutoff current	I_{CEO}	V_{CE} = 10 V, I_B = 0; $-55°C \le T_A \le +25°C$	01, 02	-	10	nA
		T_A = +125°C		-	1.0	µA
Collector to emitter cutoff current (Darlington pair) 3/	$I_{CEO(D)}$	V_{CE} = 10 V, I_B = 0; $-55°C \le T_A \le +25°C$	01	-	20	nA
		T_A = +125°C		-	50	µA
Collector to substrate cutoff current 2/	I_{CUO}	V_{CU} = 40 V; $-55°C \le T_A \le +25°C$	01, 02	-	10	nA
		T_A = +125°C		-	200	nA
Emitter to base cutoff current	I_{EBO}	V_{EB} = 4 V, I_C = 0; $-55°C \le T_A \le +25°C$	01, 02	-	10	nA
		T_A = +125°C		-	200	nA
Collector to emitter voltage (saturated)	$V_{CE(sat)}$	I_C = 10 mA, I_B = 1 mA; $-55°C \le T_A \le +25°C$	01, 02	-	.400	V
		T_A = +125°C		-	.600	V
Base emitter voltage (saturated)	$V_{BE(sat)}$	I_C = 10 mA, I_B = 1 mA; $+25°C \le T_A \le +125°C$	01, 02	-	1.0	V
		T_A = -55°C		-	1.1	V
Base emitter voltage (unsaturated)	V_{BE}	V_{CE} = 3 V, I_E = -1 mA; T_A = -55°C	01, 02	.750	.950	V
		T_A = +25°C		.600	.800	V
		T_A = +125°C		.450	.650	V
Base emitter voltage (unsaturated)	V_{BE}	V_{CE} = 3 V, I_E = -10 mA; T_A = -55°C	01, 02	-	1.000	V
		T_A = +25°C		-	.900	V
		T_A = +125°C		-	.750	V
Base emitter voltage (unsaturated), Darlington pair 3/	$V_{BE(D)}$	V_{CE} = 3 V, I_E = -1 mA; T_A = -55°C	01	1.500	1.900	V
		T_A = +25°C		1.100	1.500	V
		T_A = +125°C		.700	1.100	V
Base emitter voltage (unsaturated), Darlington pair 3/	$V_{BE(D)}$	V_{CE} = 3 V, I_E = -10 mA; T_A = -55°C	01	-	2.00	V
		T_A = +25°C		-	1.600	V
		T_A = +125°C		-	1.200	V
Input offset voltage, differential pair 5/	$\lvert V_{BEQ1} - V_{BEQ2} \rvert$	V_{CE} = 3 V, I_E = -1 mA; T_A = +25°C	01, 02	-	2.0	mV
		$-55°C \le T_A \le 125°C$		-	3.0	mV
Input offset voltage for pairs of isolated transistors 4/ 6/ 7/	$\lvert V_{BEQA} - V_{BEQB} \rvert$	V_{CE} = 3 V, I_E = -1 mA; T_A = +25°C	01, 02	-	2.0	mV
		$-55°C \le T_A \le 125°C$		-	3.0	mV

See footnotes on page 5.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions		Device type	Limits 1/ Min	Limits 1/ Max	Units
Temperature coefficient of base emitter voltage 4/ 8/	$\frac{\Delta V_{BE}}{\Delta T}$	V_{CE} = 3 V, I_E = -1 mA	-55°C≤T_A≤+25°C	01, 02	-2.2	-1.5	mV/°C
			+25°C≤T_A≤+125°C		-2.2	-1.5	mV/°C
Temperature coefficient of base emitter voltage, Darlington pair 3/ 8/	$\frac{\Delta V_{BE(D)}}{\Delta T}$	V_{CE} = 3 V, I_E = -1 mA	-55°C≤T_A≤+25°C	01	-5.0	-3.5	mV/°C
			+25°C≤T_A≤+125°C		-5.0	-3.5	mV/°C
Temperature coefficient of input offset voltage 9/ 4/	$\frac{\Delta\lvert V_{BEQA} - V_{BEQB}\rvert}{\Delta T}$	V_{CE} = 3 V, I_E = -1 mA	-55°C≤T_A≤+25°C	01, 02	-	15	µV/°C
			+25°C≤T_A≤+125°C		-	15	µV/°C
Static forward current transfer ratio (beta)	h_{FE}	V_{CE} = 3 V, I_C = 10 µA	+25°C≤T_A≤+125°C	01, 02	45	-	-
			T_A=-55°C		25	-	-
Static forward current transfer ratio (beta)	h_{FE}	V_{CE} = 3 V, I_C = 1 mA	+25°C≤T_A≤+125°C	01, 02	70	300	-
			T_A=-55°C		40	-	-
Static forward current transfer ratio (beta)	h_{FE}	V_{CE} = 3 V, I_C = 10 mA	+25°C≤T_A≤+125°C	01, 02	60	-	-
			T_A=-55°C		35	-	-
Magnitude of static beta ratio for any two isolated transistors 4/ 6/ 7/	$\frac{h_{FEQA}}{h_{FEQB}}$	V_{CE} = 3 V, I_C = 1 mA	T_A=+25°C	01, 02	.9	1.1	-
			-55°C≤T_A≤125°C		.85	1.15	-
Static forward current transfer ratio, Darlington pair 3/	$h_{FE(D)}$	V_{CE} = 3 V, I_C = 1 mA	+25°C≤T_A≤+125°C	01	4000	-	-
			T_A=-55°C		2500	-	-
Static forward current transfer ratio, Darlington pair 3/	$h_{FE(D)}$	V_{CE} = 3 V, I_C = 100 µA	+25°C≤T_A≤+125°C	01	2500	-	-
			T_A=-55°C		1500	-	-
Low frequency, small signal, forward current transfer ratio	h_{fe}	V_{CE} = 3 V, I_C = 1 mA	T_A=-55°C	01, 02	35	-	-
			+25°C≤T_A≤+125°C		60	-	-
Gain-bandwidth product	f_t	See figure 9	T_A=+25°C	01, 02	300	-	MHz
Delay time	t_d	See figure 7	T_A=-55°C	01, 02	-	160	ns
			T_A=+25°C		-	100	ns
			T_A=+125°C		-	160	ns
Rise time	t_r	See figure 7	T_A=-55°C	01, 02	-	80	ns
			T_A=+25°C		-	50	ns
			T_A=+125°C		-	80	ns
Storage time	t_s	See figure 7	T_A=-55°C	01, 02	-	300	ns
			T_A=+25°C		-	200	ns
			T_A=+125°C		-	300	ns
Fall time	t_f	See figure 7	T_A=-55°C	01, 02	-	125	ns
			T_A=+25°C		-	80	ns
			T_A=+125°C		-	125	ns
Channel separation	C.S.	See figure 8	T_A=+25°C	01, 02	80	-	dB

See footnotes on page 5.

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TABLE I FOOTNOTES

1/ Limits apply to each transistor within the array, unless otherwise specified.

2/ Does not apply to Q_5 of device type 02.

3/ Applies only to Darlington pair (Q_3, Q_4) of device type 01.

4/ Does not apply to Darlington pair (Q_3, Q_4) of device type 01.

5/ Applies only to differential pair (Q_1, Q_2) of device type 02.

6/ Does not apply to differential pair (Q_1, Q_2) of device type 02.

7/ Applies for pairs (Q_1, Q_2) of device type 01 and for pairs (Q_1, Q_3), (Q_1, Q_4), (Q_1, Q_5) of device type 02.

8/ $\dfrac{V_{BE}\ @\ 125°C - V_{BE}\ @\ 25°C}{125°C - 25°C}$, $\dfrac{V_{BE}\ @\ 25°C - V_{BE}\ @\ -55°C}{25°C - (-55°C)}$

9/ $\left|\dfrac{|V_{BEQA} - V_{BEQB}|\ @\ 125°C - |V_{BEQA} - V_{BEQB}|\ @\ 25°C}{125°C - 25°C}\right|$,

$\left|\dfrac{|V_{BEQA} - V_{BEQB}|\ @\ 25°C - |V_{BEQA} - V_{BEQB}|\ @\ -55°C}{25°C - (-55°C)}\right|$

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.6 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III and limits of table IV which constitute the minimum electrical test requirements for screening, qualification and quality conformance, by device class are specified in table II.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Class A devices	Class B devices	Class C devices
Interim electrical parameters (Pre Burn-In) (Method 5004)	1	1	None
Final electrical test parameters (Method 5004)	1*,2,3,4	1*,2,3,4	1
Group A test requirements (Method 5005)	1,2,3,4,5,6, 7,9,10,11	1,2,3,4	1,2,3,4
Groups B and C end point electrical parameters (Method 5005)	1,2,3 and table IV delta limits	Table IV delta limits and limits	Table IV delta limits and limits
Additional electrical subgroups for group C periodic inspections	None	5,6,7,9, 10,11	5,6,7,9

* PDA applies to subgroup 1 (see 4.3 e.).

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3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2. At the option of the manufacturer, the following marking may be omitted from the body of the microcircuit, but shall be retained on the initial container.

a. Country of origin.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and Method 5005 of MIL-STD-883, except as modified herein. When the accelerated test conditions (T_A = 200°C or 250°C) are used for any test on a given inspection lot, they shall be used for all pertinent quality assurance tests (screening (4.3), and group C inspection (4.4.3)) made on the lot.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, and C inspections (see 4.4.1, 4.4.2, and 4.4.3). After qualification of one or more electrically and structurally similar types with a single lead finish, other lead finishes of the same case outline may be qualified by submitting a single type in the qualified case outline to the group B, subgroup 3 test and the group C, subgroups 1, 3, and 4 tests.

4.3 Screening. Screening shall be in accordance with Method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Test samples for the group B bond strength test specified in Method 5005 of MIL-STD-883 may, at the manufacturer's option, be randomly selected immediately following the internal visual (precap) inspection and prior to sealing, except when accelerated test condition A has been selected (see 4.4.2 b.).

b. Burn-in test (Method 1015 of MIL-STD-883) 1/.

Test condition Test circuit	B Figure 4		A 2/, 3/ Figure 6	
Product assurance class	A	B	A	B
T_A minimum	125°C	125°C	250°C	200°C
t minimum	240 hours	168 hours	16 hours or 200°C 168 hours	16 hours

NOTES:

1/ The vertical columns of this table establish alternate combinations of test conditions from which the manufacturer may choose any one for a given product assurance class, at his option, unless otherwise specified in the procurement documentation. The same condition shall be used for all devices in a given inspection lot and the same condition shall be used for both burn-in (when applicable), and operating life test for any given inspection lot. Alternate 2 for stabilization bake and high temperature storage tests shall be used only when test condition A has been selected, and alternate 1 shall be used when test condition B has been selected.

2/ When accelerated test condition A is used, the centrifuge test (3.1.6) and hermeticity tests (3.1.7) of Method 5004 of MIL-STD-883 shall be performed, in that order, subsequent to the burn-in test and before the final electrical test (3.1.12) of Method 5004 of MIL-STD-883.

3/ See 4.3 a. and 4.4.2 b.

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c. Reverse bias burn-in (Method 1015 of MIL-STD-883).

Test condition	A
Test circuit	Figure 5
Product assurance class	A
T_A minimum	150°C
t minimum	72 hours

d. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

e. Percent defective allowable (PDA) - The PDA is specified as 5 percent for class A devices and 10 percent for class B devices based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with Method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

f. Stabilization bake (Method 5004 of MIL-STD-883) 1/:

	Alternate 1			Alternate 2		
Product assurance class	A	B	C	A	B	C
T_A minimum	150°C	150°C	150°C	250°C	200°C	200°C
t minimum	24 hours	24 hours	24 hours	16 hours or 200°C 168 hours	16 hours	16 hours

NOTE:

1/ The vertical columns of this table establish alternate combinations of test conditions from which the manufacturer may choose any one for a given product class, at his option, unless otherwise specified in the procurement documentation (see 4.3 b.).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of Method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of Method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein.
b. Bond strength test may be conducted on samples collected prior to sealing, except when accelerated test condition A has been selected (see 4.3 a.).

4.4.3 Group C inspection. Group C inspection shall consist of the test subgroups and LTPD values shown in table III of Method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

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b. High temperature storage test (Method 1008 of MIL-STD-883) 1/.

Product assurance class	Alternate 1			Alternate 2		
	A	B	C	A	B	C
T_A minimum	150°C	150°C	150°C	250°C	200°C	200°C
t minimum	1000 hours	1000 hours	1000 hours	100 hours or 200°C 1000 hours	100 hours	100 hours
LTPD	7	7	7	10	10	10

NOTE:
1/ The vertical columns of this table establish alternate combinations of test conditions from which the manufacturer may choose any one for a given product assurance class, at his option, unless otherwise specified in the procurement documentation (see 4.3 b.).

c. Operating life test (Method 1005 of MIL-STD-883) 1/.

Test condition	B			A 2/		
Test circuit	Figure 4			Figure 6		
Product assurance class	A	B	C	A	B	C
T_A minimum	125°C	125°C	125°C	250°C	200°C	200°C
t minimum	1000 hours	1000 hours	1000 hours	100 hours or 200°C 1000 hours	100 hours	100 hours
LTPD	5	5	5	10	10	10

NOTES:
1/ The vertical columns of this table establish alternate combinations of test conditions from which the manufacturer may choose any one for a given product assurance class, at his option, unless otherwise specified in the procurement documentation (see 4.3 b.).
2/ See 4.3 b.

d. Steady state reverse bias (Method 1005 of MIL-STD-883).

Test condition	A
Test circuit	Figure 5
Product assurance class	A
T_A minimum	150°C
t minimum	72 hours
LTPD	5

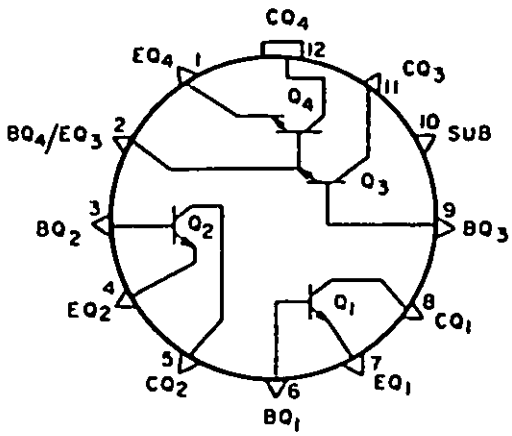
4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given are referenced to the microcircuit terminal used as ground. Currents given are conventional current and positive when flowing into the referenced terminal.

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Device type 01

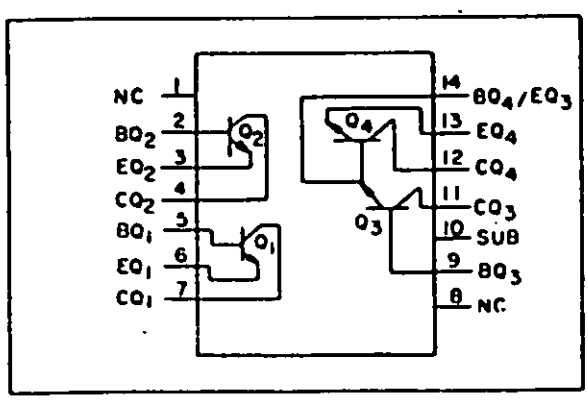
Case M



12 lead can (top view)

Device type 01

Cases A, C, D



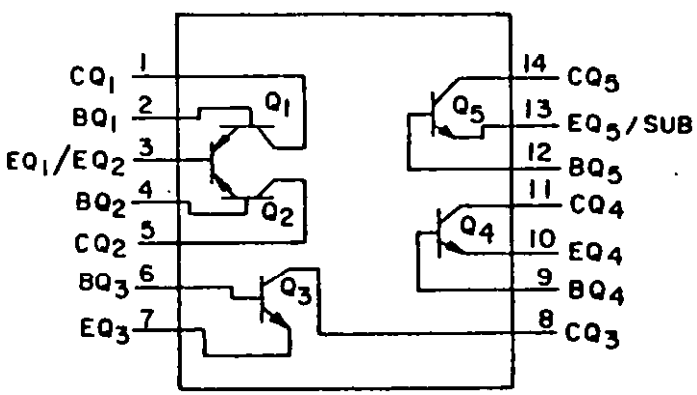
14 lead flat pack or dual-in-line (top view)

FIGURE 1. Circuit diagram and terminal connections.

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Device type 02

Cases A, C, D

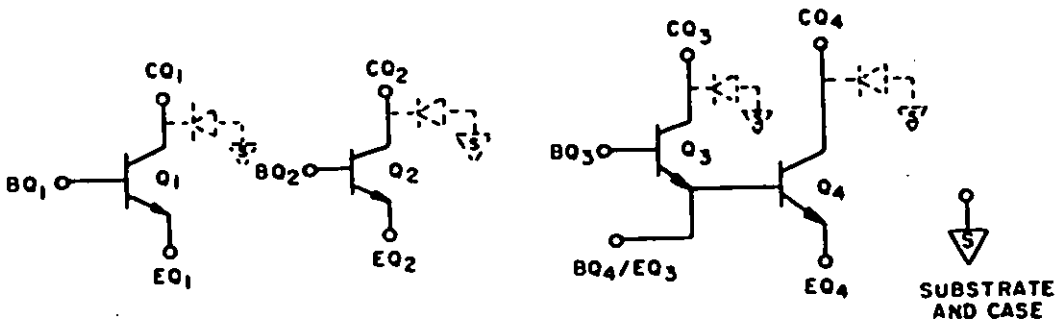


14 lead flat pack or dual-in-line (top view)

FIGURE 1. Circuit diagram and terminal connections - Continued.

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Device type 01



Device type 02

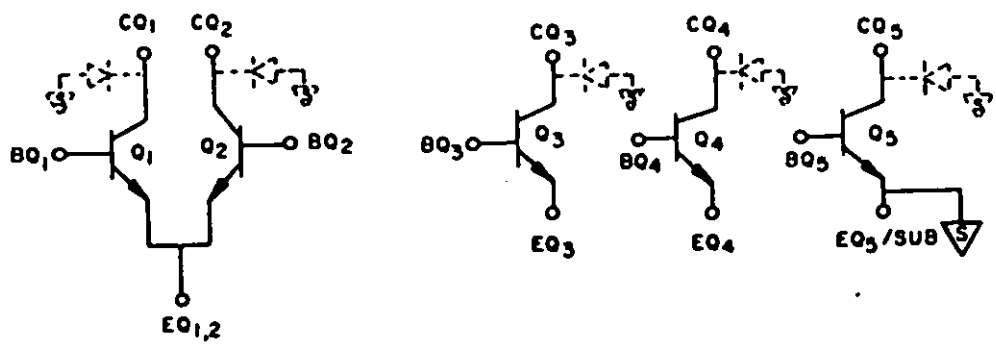
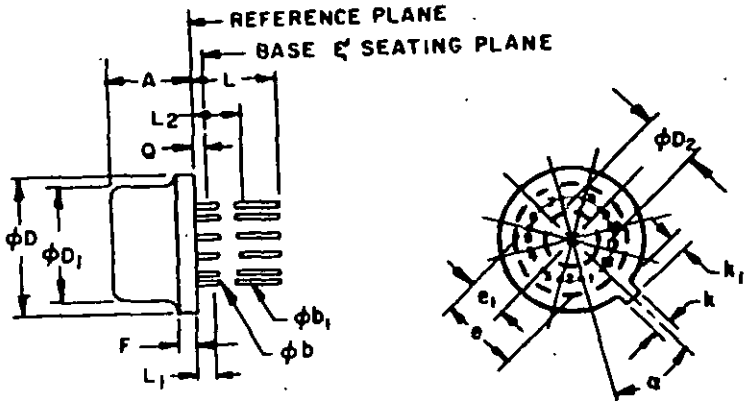


FIGURE 2. Schematic circuits.

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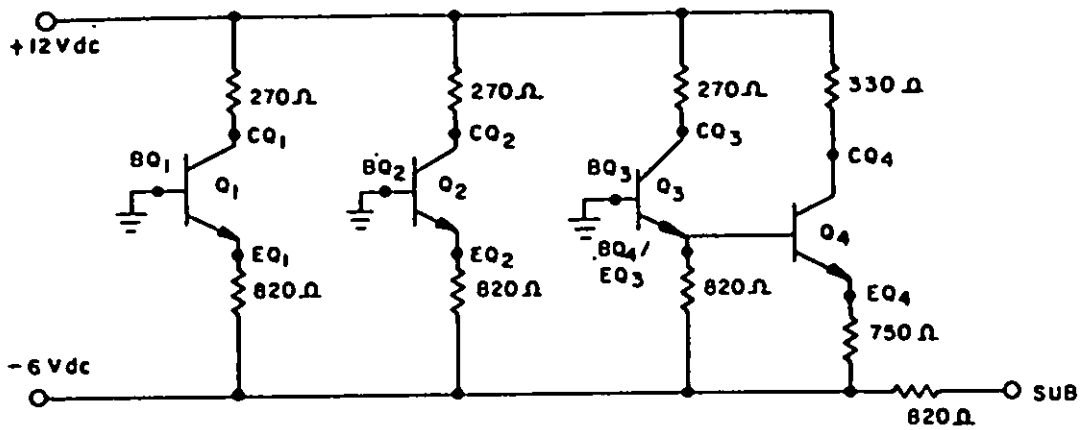
SYMBOL	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
A	.165	.185	4.19	4.70	
Øb	.016	.019	.41	.48	1
Øb1	.016	.021	.41	.53	1
ØD	.335	.370	8.51	9.40	
ØD1	.305	.335	7.75	8.51	
ØD2	.120	.160	3.05	4.06	
e	.230 BSC		5.84 BSC		3
e1	.115 BSC		2.92 BSC		3
F		.040		1.02	
k	.027	.034	.69	.86	
k1	.027	.045	.69	1.14	2
L	.500	.750	12.70	19.05	1
L1		.050		1.27	1
L2	.250		6.35		1
Q	.010	.045	.25	1.14	
α	30° BSC		30° BSC		3

NOTES:
1. (All leads) Øb applies between L1 and L2. Øb1 applies between L2 and .500 (12.70 mm) from the reference plane. Diameter is uncontrolled in L1 and beyond .500 (12.70 mm) from the reference plane.
2. Measured from the maximum diameter of the product.
3. Leads having a maximum diameter .019 (.48 mm) measured in gaging plane .054 (1.37 mm) +.001 (.03 mm) -.000 (.00 mm) below the base plane of the product shall be within .007 (.18 mm) of their true position relative to a maximum width tab.
4. The product may be measured by direct methods or by gage.

FIGURE 3. Case outline M (12-lead can).

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Device type 01



Device type 02

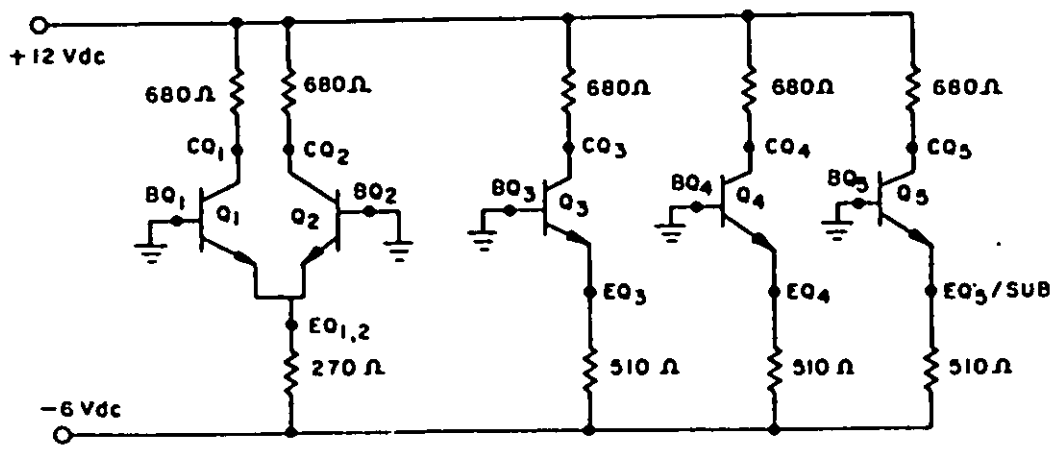
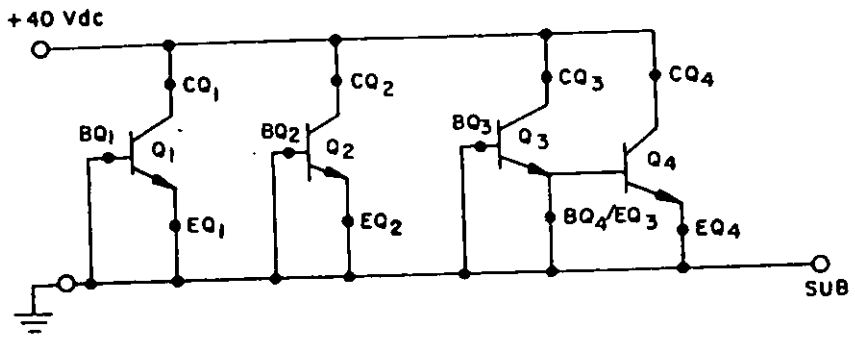


FIGURE 4. Burn-in and operating life test circuits.

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Device type 01



Device type 02

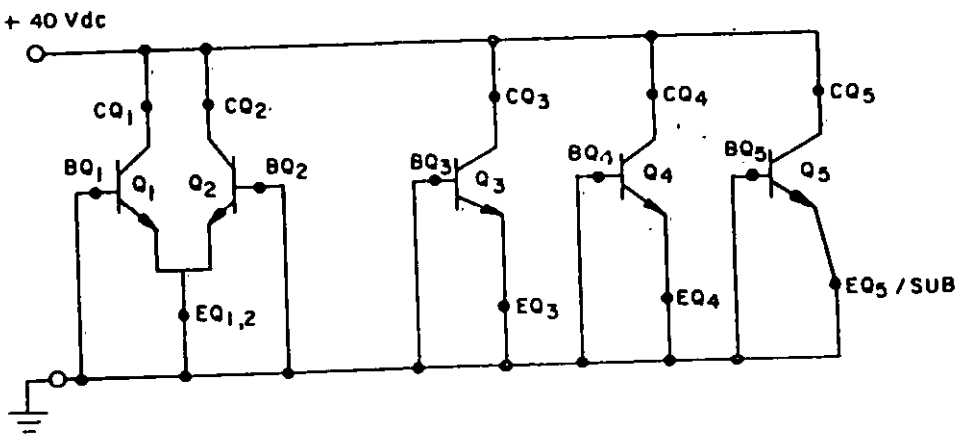
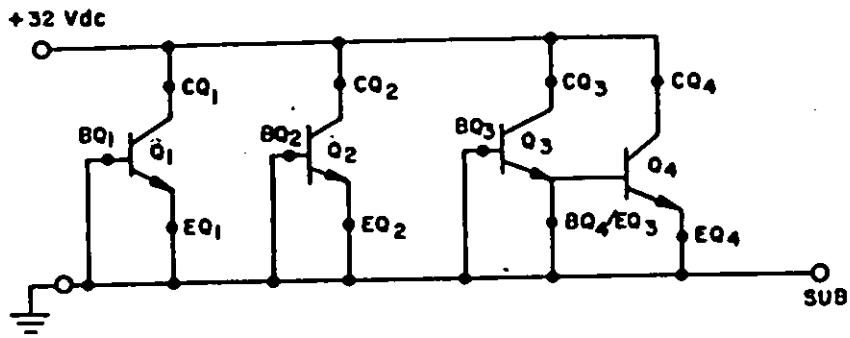


FIGURE 5. Reverse bias burn-in and life test circuits.

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Device type 01



Device type 02

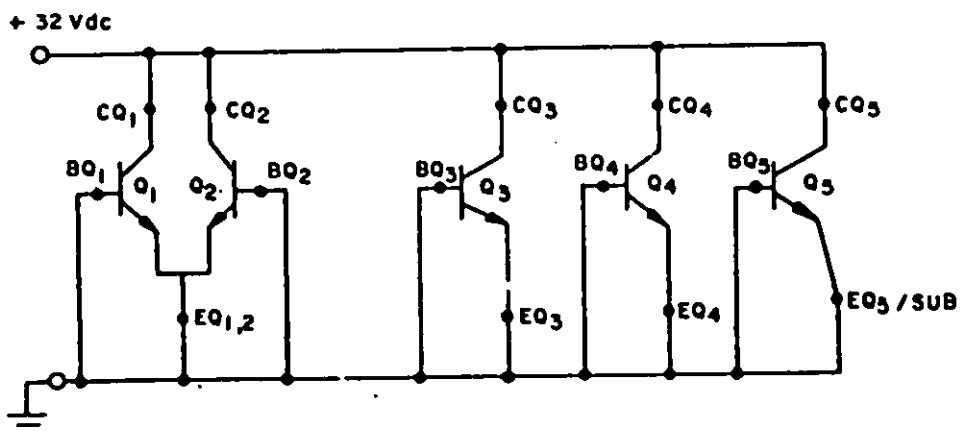


FIGURE 6. Accelerated burn-in and life test circuits.

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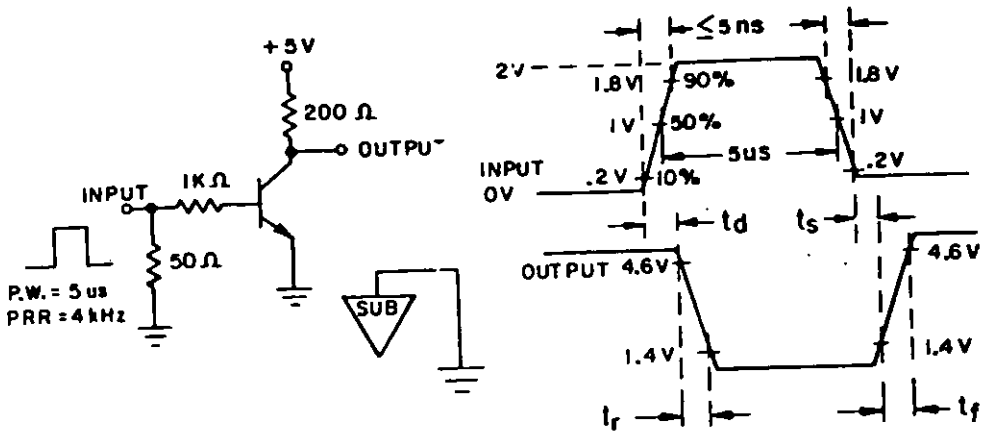
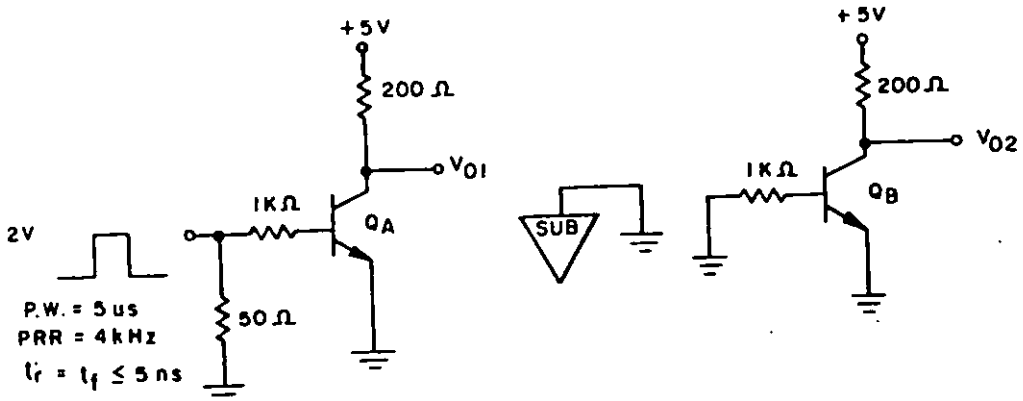


FIGURE 7. Switching time test circuit and waveforms.



NOTES:

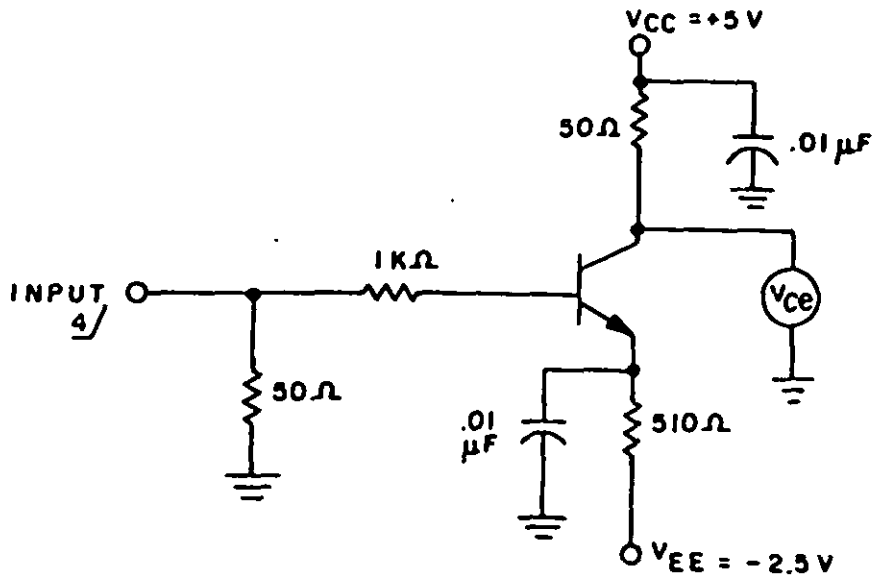
1.

Device type	Q_A	Q_B
01	Q_1	Q_2
02	Q_3	Q_4

2. Measure V_{01}, V_{02}, (volts peak)
3. Isolation = $20 \log \frac{V_{01}}{V_{02}}$

FIGURE 8. Channel separation test circuit.

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NOTES:
1. The input shall be a 100 MHz signal containing only the fundamental frequency (THD $\leq$ .5%).
2. Connect the substrate to -2.5 V.
3. With the device removed from the circuit, a shorting link is placed between the base and collector and the input signal adjusted for 1.0 mVrms on the high impedance voltmeter V_{ce}. The shorting link is then removed. The device is placed in the circuit and the reading on the voltmeter V_{ce} is equal to the magnitude of h_{fe}.
4. f_t = 100 MHz x h_{fe}.

FIGURE 9. Gain-bandwidth product (f_t) test circuit.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (Pins not designated may be H ≥ 2.0V, or L ≤ 0.8V, or open).

Subgroup	Symbol	Case A, C, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
		Case M	-	3	4	5	6	7	8	-	9	10	11	12	1	2				
		Test No.	NC	BQ2	EQ2	CQ2	BQ1	EQ1	CQ1	NC	BQ3	SUB	CQ3	CQ4	EQ4	BQ4/ EQ3		Min	Max	Unit
1	$V_{(BR)CBO}$	1					GND		10 µA			GND					CQ1	40		V
T_A = 25°C		2		GND		10 µA											CQ2			
		3									GND		10 µA				CQ3			
		4												10 µA		GND	CQ4			
	$V_{(BR)CEO}$	5						GND	1 mA								CQ1	15		
		6			GND	1 mA											CQ2			
		7											1 mA			GND	CQ3			
		8												1 mA	GND		CQ4			
	$V_{(BR)CUO}$	9							10 µA								CQ1	60		
		10				10 µA											CQ2			
		11											10 µA				CQ3			
		12												10 µA			CQ4			
	$V_{(BR)EBO}$	13					GND	10 µA									EQ1	5.0		
		14		GND	10 µA												EQ2			
		15									GND					10 µA	EQ3			
		16													10 µA	GND	EQ4			
	I_{CBO}	17					GND		35 V								CQ1		10	nA
		18		GND		35 V											CQ2			
		19									GND		35 V				CQ3			
		20												35 V		GND	CQ4			
	I_{CEO}	21						GND	10 V								CQ1			
		22			GND	10 V											CQ2			
		23											10 V			GND	CQ3			
		24												10 V	GND		CQ4			
	$I_{CEO(D)}$	25											10 V	10 V	GND		CQ3, CQ4		20	
	I_{CUO}	26							40 V								CQ1		10	
		27				40 V											CQ2			
		28											40 V				CQ3			
		29												40 V			CQ4			
	I_{EBO}	30					GND	4.0 V									EQ1			
		31		GND	4.0 V												EQ2			
		32									GND					4.0 V	EQ3			
		33													4.0 V	GND	EQ4			
	$V_{CE(sat)}$	34					1 mA	GND	10 mA								CQ1		400	V
		35		1 mA	GND	10 mA											CQ2			
		36									1 mA		10 mA			GND	CQ3			
		37												10 mA	GND	1 mA	CQ4			
	$V_{BE(sat)}$	38					1 mA	GND	10 mA								BQ1		1.0	
		39		1 mA	GND	10 mA											BQ2			
		40									1 mA		10 mA			GND	BQ3			
		41												10 mA	GND	1 mA	BQ4			
	V_{BE}	42					I_B	GND/ -1 mA	3 V								BQ1	.600	.800	
		43		I_B	GND/ -1 mA	3 V											BQ2			
		44									I_B		3 V			GND/ -1 mA	BQ3			
		45												3 V	GND/ -1 mA	I_B	BQ4			
		46					I_B	GND/ -10 mA	3 V								BQ1		900	

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (Pins not designated may be H ≥ 2.0V, or L ≤ 0.8V, or open).

Subgroup	Symbol	Case A, C, D / Case M	1 / -	2 / 3	3 / 4	4 / 5	5 / 6	6 / 7	7 / 8	8 / -	9 / 9	10 / 10	11 / 11	12 / 12	13 / 1	14 / 2	Meas. terminal	Test limits Min	Max	Unit	NOTE
		Test No.	NC	BQ2	EQ2	CQ2	BQ1	EQ1	CQ1	NC	BQ3	SUB	CQ3	CQ4	EQ4	BQ4/EQ3					
1	V_{BE}	47		I_B	GND/ -10 mA	3 V						GND					BQ2		.900	V	
T_A = 25°C		48									I_B		3 V			GND/ -10 mA	BQ3				
		49												3 V	GND/ -10 mA	I_B	BQ4				
	$V_{BE(D)}$	50									I_B		3 V	3 V	GND/ -1 mA		BQ3	1.100	1.500		
	$V_{BE(D)}$	51									I_B		3 V	3 V	GND/ -10 mA		BQ3		1.600		
	$\lvert V_{BEQ1} - V_{BEQ2} \rvert$	52															-		2.0	mV	1
	h_{FE}	53					I_B	GND	3 V/ 10 µA								BQ1	45			2
		54		I_B	GND	3 V/ 10 µA											BQ2				2
		55									I_B		3 V/ 10 µA			GND	BQ3				2
		56												3 V/10µA	GND	I_B	BQ4				2
		57					I_B	GND	3 V/ 1 mA								BQ1	70	300		3
		58		I_B	GND	3 V/ 1 mA											BQ2				3
		59									I_B		3 V/ 1 mA			GND	BQ3				3
		60												3 V/1 mA	GND	I_B	BQ4				3
		61					I_B	GND	3 V/ 10 mA								BQ1	60			4
		62		I_B	GND	3 V/ 10 mA											BQ2				4
		63									I_B		3 V/ 10 mA			GND	BQ3				4
		64												3 V/10mA	GND	I_B	BQ4				4
	h_{FEQ1}/h_{FEQ2}	65															-	0.9	1.1		5
	$h_{FE(D)}$	66									I_B		CQ4	3 V/ 100 µA	GND		BQ3	2500			6
	$h_{FE(D)}$	67									I_B		CQ4	3 V/1 mA	GND		BQ3	4000			7
2	I_{CBO}	68						GND		35 V							CQ1		.3	µA	
T_A = +125°C		69		GND		35 V											CQ2				
		70									GND		35 V				CQ3				
		71												35 V		GND	CQ4				
	I_{CEO}	72						GND	10 V								CQ1		1.0		
		73			GND	10 V											CQ2				
		74											10 V			GND	CQ3				
		75												10 V	GND		CQ4				
	$I_{CEO(D)}$	76											10 V	10 V	GND		CQ3, CQ4		50		
	I_{CUO}	77							40 V								CQ1		200	nA	
		78				40 V											CQ2				
		79											40 V				CQ3				
		80												40 V			CQ4				

NOTES:
1. Calculate $\lvert V_{43} - V_{43} \rvert$
2. Calculate h_{FE} = 10 µA/I_B
3. Calculate h_{FE} = 1 mA/I_B
4. Calculate h_{FE} = 10 mA/I_B
5. Calculate $h_{FE(57)}/h_{FE(58)}$
6. Calculate $h_{FE(D)}$ = 100 µA/I_B
7. Calculate $h_{FE(D)}$ = 1 mA/I_B

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (Pins not designated may be H ≥ 2.0V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	Case A, C, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits			NOTE
		Case M	-	3	4	5	6	7	8	-	9	10	11	12	1	2					
		Test No.	NC	BQ2	EQ2	CQ2	BQ1	EQ1	CQ1	NC	BQ3	SUB	CQ3	CQ4	EQ4	BQ4/ EQ3		Min	Max	Unit	
2 T_A = +125°C	I_{EBO}	81					GND	4.0 V				GND					EQ1		200	nA	
		82		GND	4.0 V												EQ2				
		83									GND					4.0 V	EQ3				
		84													4.0 V	GND	EQ4				
	$V_{CE(sat)}$	85					1 mA	GND	10 mA								CQ1		.600	V	
		86		1 mA	GND	10 mA											CQ2				
		87									1 mA		10 mA			GND	CQ3				
		88												10 mA	GND	1 mA	CQ4				
	V_{BE}	89					I_B	GND/ -1 mA	3 V								BQ1	.450	.650		
		90		I_B	GND/ -1 mA	3 V											BQ2				
		91									I_B		3 V			GND/ -1 mA	BQ3				
		92												3 V	GND/ -1 mA	I_B	BQ4				
		93					I_B	GND/ -10 mA	3 V								BQ1		.750		
		94		I_B	GND/ -10 mA	3 V											BQ2				
		95									I_B		3 V			GND/ -10 mA	BQ3				
		96												3 V	GND/ -10 mA	I_B	BQ4				
20	$V_{BE(D)}$	97									I_B		3 V	3 V	GND/ -1 mA		BQ3	.700	1.100		
	$V_{BE(D)}$	98									I_B		3 V	3 V	GND/ -10 mA		BQ3		1.200		
	$\lvert V_{BEQ1} - V_{BEQ2} \rvert$	99															-		3.0	mV	1
	$\Delta V_{BE}/\Delta T$	100															-	-2.2	-1.5	mV/°C	2
		101															-				3
		102															-				4
		103															-				5
	$\Delta V_{BE(D)}/\Delta T$	104															-	-5.0	-3.5		6
	$\Delta \frac{\lvert V_{BEQ1} - V_{BEQ2} \rvert}{\Delta T}$	105															-		15.0	µV/°C	7
	h_{FEQ1}	106					I_B	GND	3 V/ 1 mA								BQ1	70	300		8
	h_{FEQ2}	107		I_B	GND	3 V/ 1 mA											BQ2	70	300		8
	h_{FEQ1}/h_{FEQ2}	108															-	0.85	1.15		9
3 T_A = -55°C	$V_{BE(sat)}$	109					1 mA	GND	10 mA								BQ1		1.100	V	
		110		1 mA	GND	10 mA											BQ2				
		111									1 mA		10 mA			GND	BQ3				
		112												10 mA	GND	1 mA	BQ4				
	V_{BE}	113					I_B	GND/ -1 mA	3 V								BQ1	.750	.950		
	V_{BE}	114		I_B	GND/ -1 mA	3 V											BQ2	.750	.950		

NOTES:
1. Calculate $(V_{89} - V_{90}) \times 10^3$
2. Calculate $(V_{89} - V_{42}) \times 10^3/100$
3. Calculate $(V_{90} - V_{43}) \times 10^3/100$
4. Calculate $(V_{91} - V_{44}) \times 10^3/100$
5. Calculate $(V_{92} - V_{45}) \times 10^3/100$
6. Calculate $(V_{97} - V_{50}) \times 10^3/100$
7. Calculate $(V_{99} - V_{52}) \times 10^3/100$
8. $h_{FE} = 1$ mA/I_B
9. Calculate $h_{FE}(106)/h_{FE}(107)$

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (Pins not designated may be H ≥ 2.0V, or L ≤ 0.8V, or open).

Subgroup	Symbol	Case A, C, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits			NOTE
		Case M	-	3	4	5	6	7	8	-	9	10	11	12	1	2		Min	Max	Unit	
		Test No.	NC	BQ3	EQ2	CQ2	BQ1	EQ1	CQ1	NC	BQ3	SUB	CQ3	CQ4	EQ4	BQ4/ EQ3					
3 T_A = -55°C	V_{BE}	115									I_B	GND	3 V			GND/ -1 mA	BQ3	.750	.950	V	
		116												3 V	GND/ -1 mA	I_B	BQ4	.750	.950		
		117					I_B	GND/ -10 mA	3 V								BQ1		1.000		
		118		I_B	GND/ -10 mA	3 V											BQ2				
		119									I_B		3 V			GND/ -10 mA	BQ3				
		120												3 V	GND/ -10 mA	I_B	BQ4				
	$V_{BE(D)}$	121									I_B		3 V	3 V	GND/ -1 mA		BQ3	1.600	1.900		
	$V_{BE(D)}$	122									I_B		3 V	3 V	GND/ -10 mA		BQ3		2.000		
	$\lvert V_{BEQ1} - V_{BEQ2} \rvert$	123															-		3.0	mV	1
	$\Delta V_{BE}/\Delta T$	124															-	-2.2	-1.5	mV/°C	2
		125															-				3
		126															-				4
		127															-				5
	$\Delta V_{BE(D)}/\Delta T$	128															-	-6.0	-3.8		6
	$\Delta \lvert V_{BEQ1} - V_{BEQ2} \rvert / \Delta T$	129															-		15.0	µV/°C	7
	h_{FEQ1}	130					I_B	GND	3 V/ 1 mA								BQ1	40			8
	h_{FEQ2}	131		I_B	GND	3 V/ 1 mA											BQ2	40			8
	h_{FEQ1}/h_{FEQ2}	132															-	.85	1.15		9
	$h_{FE(D)}$	133									I_B		CQ4	3 V/1 mA	GND		BQ3	1500			10
	$h_{FE(D)}$	134									I_B		CQ4	3V/ 100 µA	GND		BQ3	1500			11
4 T_A = 25°C	h_{fe}	135					I_b	GND	3 V/ 1 mA								BQ1	60			12
	h_{fe}	136		I_b	GND	3 V/ 1 mA											BQ2	60			13
	f_t	137															CQ1	300		MHz	14
		138															CQ2				14
		139															CQ3				14
		140															CQ4				14
5 T_A = 125°C	h_{fe}	141					I_b	GND	3 V/ 1 mA			GND					BQ1	60			12
	h_{fe}	142		I_b	GND	3 V/ 1 mA											BQ2	60			13
6 T_A = -55°C	h_{fe}	143					I_b	GND	3 V/ 1 mA								BQ1	35			13
	h_{fe}	144		I_b	GND	3 V/ 1 mA											BQ2	35			13

NOTES:
1. Calculate $(V_{113} - V_{114}) \times 10^3$
2. Calculate $(V_{42} - V_{119}) \times 10^3/80$
3. Calculate $(V_{43} - V_{114}) \times 10^3/80$
4. Calculate $(V_{44} - V_{115}) \times 10^3/80$
5. Calculate $(V_{45} - V_{116}) \times 10^3/80$
6. Calculate $(V_{60} - V_{121}) \times 10^3/80$
7. Calculate $(V_{62} - V_{125}) \times 10^3/80$
8. $h_{FE} = 1$ mA/I_B
9. $h_{FE(130)}/h_{FE(131)}$
10. $h_{FE(D)} = 1$ mA/I_{BQ3}
11. $h_{FE(D)} = 100$ µA/I_{BQ3}
12. Adjust I_b until $I_C = 1.1$ mA
13. $h_{fe} = .1$ mA/ΔI_b
14. $f = 100$ MHz, measure h_{fe}. $f_t = 100\ h_{fe}$ (see figure 9)

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (Pins not designated may be $H \geq 2.0V$, or $L \leq 0.8V$, or open).

Subgroup	Symbol	Case A, C, D Case M Test No.	1 NC -	2 BQ2	3 EQ2	4 CQ2	5 BQ1	6 EQ1	7 CQ1	8 NC -	9 BQ3	10 SUB GND	11 CQ3	12 CQ4	13 EQ4	14 BQ4/EQ3	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit	NOTE 1	NOTE 2
7 T_A = +25°C	C.S.	145															CQ2	80				
9 T_A = +25°C	t_{pd}	146															CQ1		100	ns		
	t_{pd}	147															CQ2		100			
	t_r	148															CQ1		50			
	t_f	149															CQ2		50			
	t_s	150															CQ1		200			
	t_h	151															CQ2		200			
	t_w	152															CQ1		80			
	t_t	153															CQ2		80			
10 T_A = +125°C	t_{pd}	154															CQ1		160			
	t_{pd}	155															CQ2		160			
	t_r	156															CQ1		80			
	t_f	157															CQ2		80			
	t_s	158															CQ1		300			
	t_h	159															CQ2		300			
	t_w	160															CQ1		125			
	t_t	161															CQ2		125			
11 T_A = -55°C	t_{pd}	162															CQ1		160			
	t_{pd}	163															CQ2		160			
	t_r	164															CQ1		80			
	t_f	165															CQ2		80			
	t_s	166															CQ1		300			
	t_h	167															CQ2		300			
	t_w	168															CQ1		125			
	t_t	169															CQ2		125			

NOTES:
1. See figure 8.
2. See figure 7.

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TABLE III. Group A inspection for device type 02.

Terminal conditions (Pins not designated may be $H \geq 3.0V$, or $L \leq 0.8V$, or open).

Subgroup	Symbol	Case A, C, D Test No.	1 CQ1	2 BQ1	3 EQ1,2	4 BQ2	5 CQ2	6 BQ3	7 EQ3	8 CQ3	9 BQ4	10 EQ4	11 CQ4	12 BQ5	13 EQ5/SUB	14 CQ5	Meas. terminal	Test Limits Min	Test Limits Max	Test Limits Unit
1 T_A = 25°C	$V_{(BR)CBO}$	1	10 µA	GND		GND	10 µA	GND		10 µA	GND		10 µA	GND	GND	10 µA	CQ1	40		V
		2	1 mA	GND		GND	1 mA	GND		1 mA	GND		1 mA	GND	GND	1 mA	CQ2			
		3	10 µA	GND		GND	10 µA	GND		10 µA	GND		10 µA	GND	GND	10 µA	CQ3			
		4	35 V	GND		GND	35 V	GND		35 V	GND		35 V	GND	GND	35 V	CQ4			
		5	10 V	GND		GND	10 V	GND		10 V	GND		10 V	GND	GND	10 V	CQ5			
		6	40 V	GND		GND	40 V	GND		40 V	GND		40 V	GND	GND	40 V	BQ1			
		7	10 mA	GND		GND	10 mA	GND		10 mA	GND		10 mA	GND	GND	10 mA	BQ2			
		8	10 mA	GND		GND	10 mA	GND		10 mA	GND		10 mA	GND	GND	10 mA	BQ3			
	$V_{(BR)CEO}$	9	10 µA	GND	GND		10 µA	GND	GND	10 µA	GND	GND	10 µA	GND	GND	10 µA	EQ1	15		V
		10	1 mA	GND	GND		1 mA	GND	GND	1 mA	GND	GND	1 mA	GND	GND	1 mA	EQ2			
		11	10 µA	GND	GND		10 µA	GND	GND	10 µA	GND	GND	10 µA	GND	GND	10 µA	CQ1			
		12	35 V	GND	GND		35 V	GND	GND	35 V	GND	GND	35 V	GND	GND	35 V	CQ2			
		13	10 V	GND	GND		10 V	GND	GND	10 V	GND	GND	10 V	GND	GND	10 V	CQ3			
		14	40 V	GND	GND		40 V	GND	GND	40 V	GND	GND	40 V	GND	GND	40 V	CQ4			
		15	10 mA	GND	GND		10 mA	GND	GND	10 mA	GND	GND	10 mA	GND	GND	10 mA	CQ5			
		16	10 mA	GND	GND		10 mA	GND	GND	10 mA	GND	GND	10 mA	GND	GND	10 mA	BQ1			
	$V_{(BR)CUO}$	17	10 µA	GND	10 µA	GND	10 µA	GND	10 µA	10 µA	GND	10 µA	10 µA	GND	GND	10 µA	EQ1	60		V
		18	1 mA	GND	10 µA	GND	1 mA	GND	10 µA	1 mA	GND	10 µA	1 mA	GND	GND	1 mA	EQ2			
		19	10 µA	GND	10 µA	GND	10 µA	GND	10 µA	10 µA	GND	10 µA	10 µA	GND	GND	10 µA	CQ1			
		20	35 V	GND	10 µA	GND	35 V	GND	10 µA	35 V	GND	10 µA	35 V	GND	GND	35 V	CQ2			
		21	10 V	GND	10 µA	GND	10 V	GND	10 µA	10 V	GND	10 µA	10 V	GND	GND	10 V	CQ3			
		22	40 V	GND	10 µA	GND	40 V	GND	10 µA	40 V	GND	10 µA	40 V	GND	GND	40 V	CQ4			
		23	10 mA	GND	10 µA	GND	10 mA	GND	10 µA	10 mA	GND	10 µA	10 mA	GND	GND	10 mA	CQ5			
		24	10 mA	GND	10 µA	GND	10 mA	GND	10 µA	10 mA	GND	10 µA	10 mA	GND	GND	10 mA	BQ1			
	$V_{(BR)EBO}$	25	10 µA	GND	10 µA	GND	10 µA	GND	10 µA	10 µA	GND	10 µA	10 µA	GND	GND	10 µA	EQ1	5.0		V
		26	1 mA	GND	10 µA	GND	1 mA	GND	10 µA	1 mA	GND	10 µA	1 mA	GND	GND	1 mA	EQ2			
		27	10 µA	GND	10 µA	GND	10 µA	GND	10 µA	10 µA	GND	10 µA	10 µA	GND	GND	10 µA	CQ1			
		28	35 V	GND	10 µA	GND	35 V	GND	10 µA	35 V	GND	10 µA	35 V	GND	GND	35 V	CQ2			
		29	10 V	GND	10 µA	GND	10 V	GND	10 µA	10 V	GND	10 µA	10 V	GND	GND	10 V	CQ3			
		30	40 V	GND	10 µA	GND	40 V	GND	10 µA	40 V	GND	10 µA	40 V	GND	GND	40 V	CQ4			
		31	10 mA	GND	10 µA	GND	10 mA	GND	10 µA	10 mA	GND	10 µA	10 mA	GND	GND	10 mA	CQ5			
		32	10 mA	GND	10 µA	GND	10 mA	GND	10 µA	10 mA	GND	10 µA	10 mA	GND	GND	10 mA	BQ1			
	I_{CBO}	33	10 µA	GND	4 V	GND	10 µA	GND	4 V	10 µA	GND	4 V	10 µA	GND	GND	10 µA	EQ1		10	µA
		34	1 mA	GND	4 V	GND	1 mA	GND	4 V	1 mA	GND	4 V	1 mA	GND	GND	1 mA	EQ2			
		35	10 µA	GND	4 V	GND	10 µA	GND	4 V	10 µA	GND	4 V	10 µA	GND	GND	10 µA	CQ1			
		36	35 V	GND	4 V	GND	35 V	GND	4 V	35 V	GND	4 V	35 V	GND	GND	35 V	CQ2			
		37	10 V	GND	4 V	GND	10 V	GND	4 V	10 V	GND	4 V	10 V	GND	GND	10 V	CQ3			
		38	40 V	GND	4 V	GND	40 V	GND	4 V	40 V	GND	4 V	40 V	GND	GND	40 V	CQ4			
		39	10 mA	GND	4 V	GND	10 mA	GND	4 V	10 mA	GND	4 V	10 mA	GND	GND	10 mA	CQ5			
		40	10 mA	GND	4 V	GND	10 mA	GND	4 V	10 mA	GND	4 V	10 mA	GND	GND	10 mA	BQ1			
	I_{CEO}	41	10 µA	GND	GND		10 µA	GND	GND	10 µA	GND	GND	10 µA	GND	GND	10 µA	EQ1		100	µA
		42	1 mA	GND	GND		1 mA	GND	GND	1 mA	GND	GND	1 mA	GND	GND	1 mA	EQ2			
		43	10 µA	GND	GND		10 µA	GND	GND	10 µA	GND	GND	10 µA	GND	GND	10 µA	CQ1			
		44	35 V	GND	GND		35 V	GND	GND	35 V	GND	GND	35 V	GND	GND	35 V	CQ2			
		45	10 V	GND	GND		10 V	GND	GND	10 V	GND	GND	10 V	GND	GND	10 V	CQ3			
		46	40 V	GND	GND		40 V	GND	GND	40 V	GND	GND	40 V	GND	GND	40 V	CQ4			
		47	10 mA	GND	GND		10 mA	GND	GND	10 mA	GND	GND	10 mA	GND	GND	10 mA	CQ5			
		48	10 mA	GND	GND		10 mA	GND	GND	10 mA	GND	GND	10 mA	GND	GND	10 mA	BQ1			
	I_{CUO}	49	10 µA	GND	10 µA	GND	10 µA	GND	10 µA	10 µA	GND	10 µA	10 µA	GND	GND	10 µA	EQ1		1,000	µA
		50	1 mA	GND	10 µA	GND	1 mA	GND	10 µA	1 mA	GND	10 µA	1 mA	GND	GND	1 mA	EQ2			
		51	10 µA	GND	10 µA	GND	10 µA	GND	10 µA	10 µA	GND	10 µA	10 µA	GND	GND	10 µA	CQ1			
		52	35 V	GND	10 µA	GND	35 V	GND	10 µA	35 V	GND	10 µA	35 V	GND	GND	35 V	CQ2			
		53	10 V	GND	10 µA	GND	10 V	GND	10 µA	10 V	GND	10 µA	10 V	GND	GND	10 V	CQ3			
		54	40 V	GND	10 µA	GND	40 V	GND	10 µA	40 V	GND	10 µA	40 V	GND	GND	40 V	CQ4			
		55	10 mA	GND	10 µA	GND	10 mA	GND	10 µA	10 mA	GND	10 µA	10 mA	GND	GND	10 mA	CQ5			
		56	10 mA	GND	10 µA	GND	10 mA	GND	10 µA	10 mA	GND	10 µA	10 mA	GND	GND	10 mA	BQ1			
	I_{EBO}	57	10 µA	GND	10 µA	GND	10 µA	GND	10 µA	10 µA	GND	10 µA	10 µA	GND	GND	10 µA	EQ1		1,000	µA
		58	1 mA	GND	10 µA	GND	1 mA	GND	10 µA	1 mA	GND	10 µA	1 mA	GND	GND	1 mA	EQ2			
		59	10 µA	GND	10 µA	GND	10 µA	GND	10 µA	10 µA	GND	10 µA	10 µA	GND	GND	10 µA	CQ1			
		60	35 V	GND	10 µA	GND	35 V	GND	10 µA	35 V	GND	10 µA	35 V	GND	GND	35 V	CQ2			
		61	10 V	GND	10 µA	GND	10 V	GND	10 µA	10 V	GND	10 µA	10 V	GND	GND	10 V	CQ3			
		62	40 V	GND	10 µA	GND	40 V	GND	10 µA	40 V	GND	10 µA	40 V	GND	GND	40 V	CQ4			
		63	10 mA	GND	10 µA	GND	10 mA	GND	10 µA	10 mA	GND	10 µA	10 mA	GND	GND	10 mA	CQ5			
		64	10 mA	GND	10 µA	GND	10 mA	GND	10 µA	10 mA	GND	10 µA	10 mA	GND	GND	10 mA	BQ1			
	$V_{CE(sat)}$	65	10 µA	GND		GND	10 µA	GND		10 µA	GND		10 µA	GND	10 µA	10 µA	CQ1		10	V
		66	1 mA	GND		GND	1 mA	GND		1 mA	GND		1 mA	GND	10 µA	1 mA	CQ2			
		67	10 µA	GND		GND	10 µA	GND		10 µA	GND		10 µA	GND	10 µA	10 µA	CQ3			
		68	35 V	GND		GND	35 V	GND		35 V	GND		35 V	GND	10 µA	35 V	CQ4			
		69	10 V	GND		GND	10 V	GND		10 V	GND		10 V	GND	10 µA	10 V	CQ5			
		70	40 V	GND		GND	40 V	GND		40 V	GND		40 V	GND	10 µA	40 V	BQ1			
		71	10 mA	GND		GND	10 mA	GND		10 mA	GND		10 mA	GND	4 V	10 mA	BQ2			
		72	10 mA	GND		GND	10 mA	GND		10 mA	GND		10 mA	GND	GND	10 mA	BQ3			
	$V_{BE(sat)}$	73	10 µA	GND		GND	10 µA	GND		10 µA	GND		10 µA	GND	10 µA	10 µA	CQ1		10	V
		74	1 mA	GND		GND	1 mA	GND		1 mA	GND		1 mA	GND	10 µA	1 mA	CQ2			
		75	10 µA	GND		GND	10 µA	GND		10 µA	GND		10 µA	GND	10 µA	10 µA	CQ3			
		76	35 V	GND		GND	35 V	GND		35 V	GND		35 V	GND	10 µA	35 V	CQ4			
		77	10 V	GND		GND	10 V	GND		10 V	GND		10 V	GND	10 µA	10 V	CQ5			
		78	40 V	GND		GND	40 V	GND		40 V	GND		40 V	GND	10 µA	40 V	BQ1			
		79	10 mA	GND		GND	10 mA	GND		10 mA	GND		10 mA	GND	4 V	10 mA	BQ2			
		80	10 mA	GND		GND	10 mA	GND		10 mA	GND		10 mA	GND	GND	10 mA	BQ3			

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (Pins not designated may be H ≥ 2.0V, or L ≤ 0.8V, or open).

Subgroup	Symbol	Case A, C, D Test No.	1 CQ1	2 BQ1	3 EQ1, 2	4 BQ2	5 CQ2	6 BQ3	7 EQ3	8 CQ3	9 BQ4	10 EQ4	11 CQ4	12 BQ5	13 EQ5/ SUB	14 CQ5	Meas. terminal	Test limits Min	Max	Unit	NOTE
1 T_A = +25°C	V_{BE}	49	3 V	I_B	GND/ -1 mA										GND		BQ1	.600	.800	V	
		50			GND/ -1 mA	I_B	3 V										BQ2				
		51						I_B	GND/ -1 mA	3 V							BQ3				
		52									I_B	GND/ -1 mA	3 V				BQ4				
		53												I_B	GND/ -1 mA	3 V	BQ5				
		54	3 V	I_B	GND/ -10 mA										GND		BQ1		.900		
		55			GND/ -10 mA	I_B	3 V										BQ2				
		56						I_B	GND/ -10 mA	3 V							BQ3				
		57									I_B	GND/ -10 mA	3 V				BQ4				
		58												I_B	GND/ -10 mA	3 V	BQ5				
	$\lvert V_{BEQ1} - V_{BEQ2} \rvert$	59													GND		-		2.0	mV	1
	$\lvert V_{BEQ1} - V_{BEQ3} \rvert$	60															-				2
	$\lvert V_{BEQ1} - V_{BEQ4} \rvert$	61															-				3
	$\lvert V_{BEQ1} - V_{BEQ5} \rvert$	62															-				4
	h_{FE}	63	3 V/ 10 µA	I_B	GND												BQ1	45			5
		64			GND	I_B	3 V/ 10 µA										BQ2				5
		65						I_B	GND	3 V/ 10 µA							BQ3				5
		66									I_B	GND	3 V/ 10 µA				BQ4				5
		67												I_{11}		3 V/ 10 µA	BQ5				5
		68	3 V/ 1 mA	I_B	GND												BQ1	70	300		6
		69			GND	I_B	3 V/ 1 mA										BQ2				6
		70						I_B	GND	3 V/ 1 mA							BQ3				6
		71									I_B	GND	3 V/ 1 mA				BQ4				6
		72												I_B		3 V/ 1 mA	BQ5				6
		73	3 V/ 10 mA	I_B	GND												BQ1	60			7
		74			GND	I_B	3 V/ 10 mA										BQ2				7
		75						I_B	GND	3 V/ 10 mA							BQ3				7

NOTES:
1. Calculate $\lvert V_{49} - V_{50} \rvert$
2. Calculate $\lvert V_{49} - V_{51} \rvert$
3. Calculate $\lvert V_{49} - V_{52} \rvert$
4. Calculate $\lvert V_{49} - V_{53} \rvert$
5. $h_{FE} = 10\ \mu A / I_B$
6. $h_{FE} = 1\ mA / I_B$
7. $h_{FE} = 10\ mA / I_B$

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (Pins not designated may be H ≥ 2.0V, or L ≤ 0.8V, or open).

Subgroup	Symbol	Case A,C,D Test No.	1 CQ1	2 BQ1	3 EQ1, 2	4 BQ2	5 CQ2	6 BQ3	7 EQ3	8 CQ3	9 BQ4	10 EQ4	11 CQ4	12 BQ5	13 EQ5/ SUB	14 CQ5	Meas. terminal	Test limits Min	Max	Unit	NOTE
1 T_A = +25°C	h_{FE}	76									I_B	GND	3 V/ 10 mA		GND		BQ4	60			1
	h_{FE}	77												I_B		3 V/ 10 mA	BQ5	60			1
	h_{FEQ1}/h_{FEQ2}	78															-	.9	1.1		2
	h_{FEQ1}/h_{FEQ3}	79															-				3
	h_{FEQ1}/h_{FEQ4}	80															-				4
	h_{FEQ1}/h_{FEQ5}	81															-				5
2 T_A = +125°C	I_{CBO}	82	35 V	GND		GND	35 V	GND		35 V	GND		35 V	GND		35 V	CQ1		200	nA	
		83															BQ1				
		84															BQ2				
		85															BQ3				
		86															BQ4				
	I_{CEO}	87	10 V		GND		10 V		GND	10 V		GND	10 V			10 V	BQ5		1.0	μA	
		88			GND												BQ1				
		89															BQ2				
		90															BQ3				
		91															BQ4				
	I_{CUO}	92	40 V				40 V			40 V			40 V				BQ5		500	nA	
		93															BQ1		200		
		94															BQ2				
		95															BQ3				
	I_{EBO}	96		GND	4 V												BQ4				
		97			4 V	GND		GND	4 V								BQ5				
		98									GND	4 V					BQ1		.600	V	
		99															BQ2				
	$V_{CE(sat)}$	100	10 mA	1 mA	GND												BQ3				
		101			GND	1 mA	10 mA										BQ4				
		102						1 mA	GND	10 mA							BQ5				
		103									1 mA	GND	10 mA				BQ1				
		104												1 mA		10 mA	BQ2				
	V_{BE}	105	3 V	I_B	GND/ -1 mA												BQ3	.450	.650		
		106			GND/ -1 mA	I_B	3 V										BQ4				
		107						I_B	GND/ -1 mA	3 V							BQ5				
		108									I_B	GND/ -1 mA	3 V				BQ1				
		109												I_B	GND/ -1 mA GND	3 V	BQ2				
		110	3 V	I_B	GND/ -10 mA												BQ3		.750		
		111			GND/ -10 mA	I_B	3 V										BQ4				
		112						I_B	GND/ -10 mA	3 V							BQ5				
		113									I_B	GND/ -10 mA	3 V				BQ1				

NOTES:
1. h_{FE} = 10 mA/I_B
2. $h_{FE}(68)/h_{FE}(69)$
3. $h_{FE}(68)/h_{FE}(70)$
4. $h_{FE}(68)/h_{FE}(71)$
5. $h_{FE}(68)/h_{FE}(72)$

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TABLE III. Group A inspection for device type 02 - Continued.

Terminal conditions (Pins not designated may be H ≥ 2.0V, or L ≤ 0.8V, or open).

Subgroup	Symbol	Cases X, C, D Test No.	1 CQ1	2 BQ1	3 EQ1, 2	4 BQ2	5 CQ2	6 BQ3	7 EQ3	8 CQ3	9 BQ4	10 EQ4	11 CQ4	12 BQ5	13 EQ5/ SUB	14 CQ5	Meas. terminal	Test limits Min	Max	Unit	NOTE
2 T_A = 125°C	V_{BE}	114												I_B	GND/ -10 mA GND	3 V	BQ5		.750	V	
	$\lvert V_{BEQ1} - V_{BEQ2} \rvert$	115															-		3.0	mV	1
	$\lvert V_{BEQ1} - V_{BEQ3} \rvert$	116															-				2
	$\lvert V_{BEQ1} - V_{BEQ4} \rvert$	117															-				3
	$\lvert V_{BEQ1} - V_{BEQ5} \rvert$	118															-				4
	$\frac{\Delta V_{BE}}{\Delta T}$	119															-	-2.2	-1.5	mV/°C	5
		120															-				6
		121															-				7
		122															-				8
		123															-				9
	$\frac{\Delta \lvert V_{BEQ1} - V_{BEQ2} \rvert}{\Delta T}$	124															-		15.0	µV/°C	10
	$\frac{\Delta \lvert V_{BEQ1} - V_{BEQ3} \rvert}{\Delta T}$	125															-				11
	$\frac{\Delta \lvert V_{BEQ1} - V_{BEQ4} \rvert}{\Delta T}$	126															-				12
	$\frac{\Delta \lvert V_{BEQ1} - V_{BEQ5} \rvert}{\Delta T}$	127															-				13
	h_{FE}	128	3 V/ 1 mA	I_B	GND												BQ1	70	300		14
		129			GND	I_B	3 V/ 1 mA										BQ2				14
		130						I_B	GND	3 V/ 1 mA							BQ3				14
		131									I_B	GND	3 V/ 1 mA				BQ4				14
		132												I_B		3 V/ 1 mA	BQ5				14
	h_{FEQ1}/h_{FEQ2}	133															-	.85	1.15		15
	h_{FEQ1}/h_{FEQ3}	134															-				16
	h_{FEQ1}/h_{FEQ4}	135															-				17
	h_{FEQ1}/h_{FEQ5}	136															-				18
3 T_A = -55°C	$V_{BE(sat)}$	137	10 mA	1 mA	GND	1 mA	10 mA	1 mA	GND	10 mA	1 mA	GND	10 mA	1 mA		10 mA	BQ1		1.100	V	
		138			GND												BQ2				
		139															BQ3				
		140															BQ4				
		141															BQ5				
	V_{BE}	142	3 V	I_B	GND/ -1 mA	I_B	3 V										BQ1	.750	.950		
		143			GND/ -1 mA												BQ2				
		144						I_B	GND/ -1 mA	3 V							BQ3				
		145									I_B	GND/ -1 mA	3 V				BQ4				
		146												I_B	GND/ -1 mA	3 V	BQ5				

NOTES:

1. Calculate $\lvert V_{105} - V_{106} \rvert$
2. Calculate $\lvert V_{105} - V_{107} \rvert$
3. Calculate $\lvert V_{105} - V_{108} \rvert$
4. Calculate $\lvert V_{105} - V_{109} \rvert$
5. Calculate $V_{105} - V_{49}/100$
6. Calculate $V_{106} - V_{50}/100$
7. Calculate $V_{107} - V_{51}/100$
8. Calculate $V_{108} - V_{52}/100$
9. Calculate $V_{109} - V_{53}/100$
10. Calculate $\lvert V_{115} - V_{59} \rvert /100$
11. Calculate $\lvert V_{116} - V_{60} \rvert /100$
12. Calculate $\lvert V_{117} - V_{61} \rvert /100$
13. Calculate $\lvert V_{118} - V_{62} \rvert /100$
14. h_{FE} = 1 mA/I_B
15. Calculate $h_{FE(128)}/h_{FE(129)}$
16. Calculate $h_{FE(128)}/h_{FE(130)}$
17. Calculate $h_{FE(128)}/h_{FE(131)}$
18. Calculate $h_{FE(128)}/h_{FE(132)}$

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (Pins not designated may be H ≥ 2.0V, or L ≤ 0.8V, or open).

Subgroup	Symbol	Case A, C, D Test No.	1 CQ1	2 BQ1	3 EQ1, 2	4 BQ2	5 CQ2	6 BQ3	7 EQ3	8 CQ3	9 BQ4	10 EQ4	11 CQ4	12 BQ5	13 EQ5/ SUB	14 CQ5	Meas. terminal	Test limits Min	Max	Unit	NOTE
3 T_A = -55°C	V_{BE}	147	3 V	I_B	GND/ -10 mA										GND		BQ1		1.000	V	
		148			GND/ -10 mA	I_B	3 V										BQ2				
		149						I_B	GND/ -10 mA	3 V							BQ3				
		150									I_B	GND/ -10 mA	3 V				BQ4				
		151												I_B	GND/ -10 mA	3 V	BQ5				
	$\lvert V_{BEQ1} - V_{BEQ2} \rvert$	152													GND		-		3.0	mV	1
	$\lvert V_{BEQ1} - V_{BEQ3} \rvert$	153															-				2
	$\lvert V_{BEQ1} - V_{BEQ4} \rvert$	154															-				3
	$\lvert V_{BEQ1} - V_{BEQ5} \rvert$	155															-				4
	$\Delta V_{BE}/\Delta T$	156															-	-2.2	-1.5	mV/°C	5
		157															-				6
		158															-				7
		159															-				8
		160															-				9
	$\Delta \lvert V_{BEQ1} - V_{BEQ2} \rvert / \Delta T$	161															-		15.0	µV/°C	10
	$\Delta \lvert V_{BEQ1} - V_{BEQ3} \rvert / \Delta T$	162															-				11
	$\Delta \lvert V_{BEQ1} - V_{BEQ4} \rvert / \Delta T$	163															-				12
	$\Delta \lvert V_{BEQ1} - V_{BEQ5} \rvert / \Delta T$	164															-				13
	h_{FE}	165	3 V/ 1 mA	I_B	GND												BQ1	40			14
		166			GND	I_B	3 V/ 1 mA										BQ2				14
		167						I_B	GND	3 V/ 1 mA							BQ3				14
		168									I_B	GND	3 V/ 1 mA				BQ4				14
		169												I_B		3 V/ 1 mA	BQ5				14
	h_{FEQ1}/h_{FEQ2}	170															-	.83	1.15		15
	h_{FEQ1}/h_{FEQ3}	171															-				16
	h_{FEQ1}/h_{FEQ4}	172															-				17
	h_{FEQ1}/h_{FEQ5}	173															-				18
4 T_A = +25°C	h_{fe}	174	3V/ 1 mA	I_b	GND												BQ1	60			19
	h_{fe}	175			GND	I_b	3 V/ 1 mA										BQ2	60			19
	h_{fe}	176						I_b	GND	3 V/ 1 mA							BQ3	60			19

NOTES:
1. Calculate $\lvert V_{142} - V_{143} \rvert \times 10^3$
2. Calculate $\lvert V_{142} - V_{144} \rvert \times 10^3$
3. Calculate $\lvert V_{142} - V_{145} \rvert \times 10^3$
4. Calculate $\lvert V_{142} - V_{146} \rvert \times 10^3$
5. Calculate $(V_{49} - V_{142}) \times 10^3/80$
6. Calculate $(V_{50} - V_{143}) \times 10^3/80$
7. Calculate $(V_{51} - V_{144}) \times 10^3/80$
8. Calculate $(V_{52} - V_{145}) \times 10^3/80$
9. Calculate $(V_{53} - V_{146}) \times 10^3/80$
10. Calculate $\lvert V_{59} - V_{152} \rvert \times 10^3/80$
11. Calculate $\lvert V_{60} - V_{153} \rvert \times 10^3/80$
12. Calculate $\lvert V_{61} - V_{154} \rvert \times 10^3/80$
13. Calculate $\lvert V_{62} - V_{155} \rvert \times 10^3/80$
14. h_{FE} = 1 mA/I_B
15. Calculate $h_{FE}(165)/h_{FE}(166)$
16. Calculate $h_{FE}(165)/h_{FE}(167)$
17. Calculate $h_{FE}(165)/h_{FE}(168)$
18. Calculate $h_{FE}(165)/h_{FE}(169)$
19. Adjust I_b until I_C = 1.1 mA, h_{fe} = .1 mA/ΔI_b

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (Pins not designated may be H ≥ 2.0V, or L ≤ 0.8V, or open).

Subgroup	Symbol	Case A, C, D Test No.	1 CQ1	2 BQ1	3 EQ1, 2	4 BQ2	5 CQ2	6 BQ3	7 EQ3	8 CQ3	9 BQ4	10 EQ4	11 CQ4	12 BQ5	13 EQ5/ SUB	14 CQ5	Meas. terminal	Test limits Min	Max	Unit	NOTE
4 T_A = +25°C	h_{fe}	177									I_b	GND	3 V/ 1 mA		GND		BQ4	60			1
	h_{fe}	178												I_b	GND	3 V/ 1 mA	BQ5	60			1
	f_t	179															CQ1	300		MHz	2
		180															CQ2				2
		181															CQ3				2
		182															CQ4				2
		183															CQ5				2
5 T_A = +125°C	h_{fe}	184	3 V/ 1 mA	I_b	GND										GND		BQ1	60			1
		185			GND	I_b	3 V/ 1 mA										BQ2				
		186						I_b	GND	3 V/ 1 mA							BQ3				
		187									I_b	GND	3 V/ 1 mA				BQ4				
		188												I_b		3 V/ 1 mA	BQ5				
6 T_A = -55°C	h_{fe}	189	3 V/ 1 mA	I_b	GND												BQ1	35			
		190			GND	I_b	3 V/ 1 mA										BQ2				
		191						I_b	GND	3 V/ 1 mA							BQ3				
		192									I_b	GND	3 V/ 1 mA				BQ4				
		193												I_b		3 V/ 1 mA	BQ5				
7 T_A = +25°C	C.S.	194															CQ4	80		dB	3
9 T_A = +25°C	t_d	195															CQ1		100	ns	4
		196															CQ2				
		197															CQ3				
		198															CQ4				
		199															CQ5				
	t_r	200															CQ1		50		
		201															CQ2				
		202															CQ3				
		203															CQ4				
		204															CQ5				
	t_s	205															CQ1		200		
		206															CQ2				
		207															CQ3				
		208															CQ4				
		209															CQ5				
	t_f	210															CQ1		80		
		211															CQ2				
		212															CQ3				
		213															CQ4				
		214															CQ5				

NOTES:
1. Adjust I_b until I_C = 1.1 mA. h_{fe} = 1 mA, = I_b.
2. f = 100 MHz, measure h_{fe}. f_t = 100 h_{fe} (see figure 9)
3. See figure 8
4. See figure 7

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TABLE III. Group A inspection for device type 02 - Continued.

Terminal conditions (Pins not designated may be H ≥ 3.0V, or L ≤ 0.8V, or open)

Subgroup	Symbol	Case A, C, D Test No.	1 CQ1	2 DQ1	3 EQ1, 2	4 BQ2	5 CQ2	6 BQ3	7 EQ3	8 CQ3	9 BQ4	10 EQ4	11 CQ4	12 BQ5	13 EQ5/EUB	14 CQ5	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit	NOTE
10 T_A = 125°C	t_0 → t_1 → t_2 → t_3	215													GND		Q1		160	ns	1
		216															Q2		80		
		217															Q3		300		
		218															Q4		125		
		219															Q5				
		220															Q6				
		221															Q7				
		222															Q8				
		223															Q9				
		224															Q10				
		225															Q11				
		226															Q12				
		227															Q13				
		228															Q14				
		229															Q15				
		230															Q16				
		231															Q17				
		232															Q18				
		233															Q19				
		234															Q20				
11 T_A = -55°C	t_0 → t_1 → t_2 → t_3	235													GND		Q1		160	ns	1
		236															Q2		80		
		237															Q3		300		
		238															Q4		125		
		239															Q5				
		240															Q6				
		241															Q7				
		242															Q8				
		243															Q9				
		244															Q10				
		245															Q11				
		246															Q12				
		247															Q13				
		248															Q14				
		249															Q15				
		250															Q16				
		251															Q17				
		252															Q18				
		253															Q19				
		254															Q20				

NOTE:
1. See figure 7

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4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias.

TABLE IV. Groups B and C end point electrical parameters (T_A = 25°C).

Test	Device types 01 and 02	
	Limit	Delta
V_{BE}	.600 V min .800 V max	±.010 V
$\lvert V_{BEQA} - V_{BEQB} \rvert$	2.0 mV	
h_{FE}	70 min 300 max	±10%

4.6 Packaging inspection. The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with the requirements of MIL-M-38510, except that the rough handling test shall not apply.

5. PACKAGING

5.1 Packaging requirements. The requirements for packing shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.3 Ordering data. Procurement documents should specify the following:

a. Complete part number (see 1.2).
b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
c. Requirement for certificate of compliance, if applicable.
d. Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.
e. Requirements for packaging and packing.
f. Requirements for failure analysis (including required test condition of Method 5003), corrective action and reporting of results, if applicable.
g. Requirements for product assurance options.
h. Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-STD-1313, MIL-STD-1331, MIL-S-19500E (Appendix B), and as follows:

$V_{(BR)CUO}$ - - - Breakdown voltage, collector to substrate
I_{CUO} - - - - - Collector to substrate cutoff current
V_{CU} - - - - - Collector to substrate voltage (dc).

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6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. Microcircuits covered by this specification are substitutable for the following commercial device types:

Device type	Commercial type
01	3018A
02	3045

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 80, 85

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DSA - ES

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